

Lin Li

CONTACT & SOCIAL MEDIAS

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ACADEMIC APPOINTMENTS

Postdoctoral Researcher, University of Oxford 2025 – Present

[Oxford Applied and Theoretical Machine Learning \(OATML\)](#) Group, Department of CS, UK

- Advisor: [Prof. Yarin Gal](#).
- Responsibilities: lead research projects; support group management and teaching activities; supervise MSc students; funding acquisition.

Pesearch Fellow, University of Birmingham 2025 – Present

School of Computer Science, UK

- Work with [Prof. Huiping Chen](#).
- Responsibilities: lead the research project; supervise MSc student; initiate collaboration.

Research Intern, Tencent 2021 – 22

[Robotics X Lab](#), Shenzhen, China

- Mentor: [Dr. Lipeng Chen](#).
- Responsibilities: co-led the project on enabling robots to learn throwing and catching skills.

EDUCATION

PhD in Machine Learning, King's College London 2019 – 24

Department of Informatics, London, UK

- Advisors: [Prof. Michael Spratling](#) (primary) and [Prof. Dimitrios Letsios](#).
- Thesis: *Towards Robust Visual Classification through Adversarial Training*.
- Thesis Committee: [Prof. George D. Magoulas](#) and [Dr. Adel Bibi](#).

MSc in Computing Science, Imperial College London 2017 – 18

Department of Computing, London, UK

- Advisor: [Prof. Wayne Luk](#).
- Grade: Overall Distinction (Exam + Thesis).
- Dissertation: *Understanding Deep CNNs via Interpretable Individual Units*.

Department of Finance, School of Management, Xiamen, China

- Advisor: [Prof. Zheng Qiao](#).
- Grade: top 10% of graduating class.
- Dissertation: *Quantitatively Measuring Investor's Sentiment via Search Index*.

PUBLICATIONS

(* indicates equal contribution; † indicates the corresponding author if have)

AI Safety:

1. **Lin Li**†, Michael Spratling, "[Robust Shortcut and Disordered Robustness: Improving Adversarial Training Through Adaptive Smoothing](#)", Pattern Recognition (**PR**), 2025
2. **Lin Li**, Yifei Wang, Chawin Sitawarin, Michael Spratling, "[OODRobustBench: a Benchmark and Large-Scale Analysis of Adversarial Robustness under Distribution Shift](#)", International Conference on Machine Learning (**ICML**), 2024
3. **Lin Li***, Haoyan Guan*, Jianing Qiu, Michael Spratling, "[One Prompt Word is Enough to Boost Adversarial Robustness for Pre-trained Vision-Language Models](#)", IEEE/CVF Computer Vision and Pattern Recognition Conference (**CVPR**), 2024
4. **Lin Li**†, Jianing Qiu, Michael Spratling, "[AROID: Improving Adversarial Robustness Through Online Instance-Wise Data Augmentation](#)", the International Journal of Computer Vision (**IJCV**), 2024
5. **Lin Li**, Michael Spratling, "[Data augmentation alone can improve adversarial training](#)", International Conference on Learning Representations (**ICLR**), 2023
6. **Lin Li**, Michael Spratling†, "[Understanding and combating robust overfitting via input loss landscape analysis and regularization](#)", Pattern Recognition (**PR**), 2023
7. Jahan C Penny-Dimri, Magdalena Bachmann, William R Cooke, Sam Mathewlynn, Samuel Dockree, John Tolladay, Jannik Kossen, **Lin Li**, Yarin Gal, Gabriel Davis Jones, "[Reducing Large Language Model Safety Risks in Women's Health using Semantic Entropy](#)", in submission, 2025
8. Jianing Qiu, **Lin Li**, Jiankai Sun, Hao Wei, Zhe Xu, Kyle Lam, Wu Yuan, "[Emerging Cyber Attack Risks of Medical AI Agents](#)", in submission, 2025

AI + Healthcare / Robotics:

9. Jianing Qiu, **Lin Li**, Jiankai Sun, and Jiachuan Peng, Peilun Shi, Ruiyang Zhang, Yinzhaohong, Kyle Lam, Frank P.-W. Lo, Bo Xiao, Wu Yuan, Dong Xu†, Benny Lo†, "[Large AI Models in Health Informatics: Applications, Challenges, and the Future](#)", IEEE Journal of Biomedical and Health Informatics (**JBHI**), 2023
10. Lipeng Chen*†, Jianing Qiu*, **Lin Li***, Xi Luo, Guoyi Chi, Yu Zheng, "[Advancing Robots with Greater Dynamic Dexterity: A Large-Scale Multi-View and Multi-Modal Dataset of Human-Human Throw&Catch of Arbitrary Objects](#)", International Journal of Robotics Research (**IJRR**), 2024
11. Jianing Qiu*, Jian Wu*, Hao Wei*, and Peilun Shi, Mingqing Zhang, Yunyun Sun, **Lin Li**, ... (+34 authors), Wu Yuan†, "[Development and Validation of a Multimodal Multitask Vision Foundation Model for Generalist Ophthalmic Artificial Intelligence](#)", New England Journal of Medicine Artificial

Intelligence (NEJM AI), 2024

12. Linyuan Li, Anujit Saha, **Lin Li**, Boyuan Li, Mengxian He, Jianing Qiu, Wu Yuan, "[Artificial Intelligence for Biomedical Video Generatio](#)", in submission, 2024

SOFTWARES & OPEN-SOURCE

OODRobustBench 2024

- Description: the first benchmark for evaluating adversarial robustness under distribution shifts.
- Contribution: led the project; developed the concept, code, and leaderboards.
- [Leaderboards](#), [Evaluation Toolbox](#).

Human-Human Throw&Catch (H2TC) Dataset 2024

- Description: the first large multimodal dataset of human throw-catch actions for humanoid robots.
- Contribution: co-led the project; co-developed the concept, code, and dataset.
- [Website \(Data, Tools and Tutorials\)](#).

Awesome-Healthcare-Foundation-Model 2023

- Description: a continual paper collection of foundation models for healthcare (46 folks and 450 stars).
- Contribution: co-created the collection; co-authored a survey based on it.
- [Paper collection](#).

Adversarial Robustness Toolbox (ART), Linux Foundation AI & Data Foundation 2020

- Description: a popular (1.2k+ folks and 5.2k+ stars on Github) [library](#) for adversarial ML research.
- Contribution: fix the bugs as recognized [here](#).

Smart Acc, Microsoft 2018

- Description: a web application based on object detection algorithms for accomodation search.
- Contribution: co-led the project; developed the core object detection algorithms.
- [Blog \(Microsoft\)](#), [Code](#), [Report](#), [Demo](#).

GRANTS

Horizon Europe (€29M) 2025-27

Proposal Contributor, Research Project Lead

- Description: the EU's key funding programme for research and innovation.
- Proposal Title: Diversis Viis, Plurima Solvens (DVPS)
- Responsibilities: co-developed the concept of multimodal foundation models for human and society; wrote the corresponding sections in the proposal; lead research on multimodal hallucination.

IDAI Pump Prime (£20K) 2025

Key Personnel (named on the proposal), Research Project Lead

- Description: a priming fund from The Institute for Data and AI (IDAI), University of Birmingham.
- Proposal Title: Proactive defense against Deepfake
- Responsibilities: co-lead the project with [Prof. Huiping Chen](#); direct the research; co-supervise MSc.

AWARDS & HONORS

OpenAI API Access (Credits \$300) , OpenAI and Oxford	2025
Nomination of USERN Prize , USERN	2025
King's-China Scholarship (£109.8K) , KCL and CSC	2019-24
PGR Research Support (£2.5K) , KCL	2023
1st Class (Xiangyu) University Scholarship (¥7K) , Xiamen University	2016
Excellent Academic Performance Scholarship (¥1K) , Xiamen University	2015
3rd prize winner (¥5K) , Jinyuan Creativity and Startup Contest, Xiamen City	2016
3rd prize winner , ChinaNet Dream Accelerator Programming Contest, China	2015

TEACHING

Teaching Assistant	2025
Department of Computer Science, University of Oxford, Oxford, UK	
• Course: Uncertainty in Deep Learning. • Responsibilities: marking individual projects.	
Teaching Assistant	2021
Department of Informatics, King's College London, London, UK	
• Courses: Machine Learning and Pattern Recognition, Introduction to Artificial Intelligence. • Responsibilities: small group tutorial, lab sessions, marking courseworks. I also co-developed a tool for automatically marking individual projects of the course, which remains in use to this day.	

INTERVIEW & PRESS MENTION

Süddeutsche Zeitung (KZ)	2025
Leading national newspapers, German	
• Title: Are Language Models Becoming Even More Unreliable? • Activity: interviewed by Nils Althaus to speak on the hallucination of latest reasoning models.	
South China Morning Post (SCMP)	2025
Newspapers, Hong Kong, China	
• Title: DeepSeek's AI in hospitals is 'too fast, too soon', Chinese medical researchers warn • Activity: Reference our investigation into emerging cyber attacks targeting medical AI agents.	
South China Morning Post (SCMP)	2024
Newspapers, Hong Kong, China	

- Title: [A Hong Kong AI model proves more accurate than doctors in diagnosing eye conditions](#)
- Activity: A report detailing our research on Ophthalmic Vision Foundation Models.

INVITED TALKS

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| AI Agents for Surgery Workshop | 2025 |
| 17th Hamlyn Symposium on Medical Robotics , UK | |
| <ul style="list-style-type: none"> • Title: Hallucination of AI Agents. | |
| CHI Seminar | 2025 |
| Lab internal seminar, Computational Health Informatics (CHI) Lab , University of Oxford, UK | |
| <ul style="list-style-type: none"> • Title: Prompting VLMs for accuracy and robustness | |
| Nanqiang Youth Scholar Forum | 2024 |
| Internatioanl faculty recruitment event, Xiamen University, China. | |
| <ul style="list-style-type: none"> • Title: Towards Safe Multimodal Foundational Models in Finance. | |
| Vision And Learning Seminar (Valse) | 2024 |
| National ML/CV seminar (5k+ attendees), China. | |
| <ul style="list-style-type: none"> • Title: One Prompt Word is Enough to Boost Adversarial Robustness for Pre-trained VLMs. | |
| AI Time Youth PhD Talk | 2023 |
| Online AI forum, Tsinghua University, China. | |
| <ul style="list-style-type: none"> • Title: Data augmentation can improve adversarial training. | |
| Departmental Research Showcase | 2023 |
| Research showcase to the public, KCL, UK. | |
| <ul style="list-style-type: none"> • Title: Defending DNNs against adversarial examples. | |

ACADEMIC SERVICE

Reviewer

- Conferences: NeurIPS, ICML, ICLR, AAAI, ECCV.
- Journals: IEEE T-Pattern Analysis and Machine Intelligence (T-PAMI), Pattern Recognition (PR), IEEE T-Information Forensics & Security (T-IFS), IEEE T-Dependable and Secure Computing (T-DSC), IEEE Journal of Biomedical and Health Informatics (J-BHI).

Program Committee

- Workshops: [2nd MEIS Workshop@CVPR2025](#), [WIHR@ICRA 2024](#), [Workshop on Cooperative Intelligence for Embodied AI@ECCV2024](#)

ACADEMIC REFEREES

(refer to their websites for the latest information)

Yarin Gal

Associate Professor of ML, Department of Computer Science, University of Oxford.

Turing AI Fellow, The Alan Turing Institute, London, UK.

Director of Research, UK AI Security Institute (AISi), London, UK.

Tutorial Fellow, Christ Church College, University of Oxford.

Phone: +44 1865 610497, Email: yarin@cs.ox.ac.uk.

Web: www.cs.ox.ac.uk/people/yarin.gal/website

Michael Spratling

Researcher, Department of Behavioural and Cognitive Sciences, University of Luxembourg.

was Reader¹ in Computational Neuroscience and Visual Cognition, Department of Informatics, KCL.

Phone: +352 46 66 44 5722, Email: michael.spratling@uni.lu.

Web: mwspratling.codeberg.page

Adel Bibi

Senior Researcher (G9: equ. spinal point to Associate Professor) in ML, Department of Engineering Science, University of Oxford.

Research Member of the Common Room at Kellogg College, University of Oxford.

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Web: www.adelbibi.com

¹ Michael held this position while serving as my PhD supervisor.